

Tyler Luo

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EDUCATION

Georgia Institute of Technology

Jan 2023 – May 2026

B.S. Computer Engineering (Cybersecurity & Information Internetworks Threads)

Atlanta, GA

Relevant Coursework: Information Security, Database Systems, Data Structures, Object Oriented Programming, Algorithm Analysis, Circuit Analysis, Cryptography for Embedded Systems, Digital Design Lab, Computer Networking

EXPERIENCE

Software Developer Intern

May 2026 – Aug 2026

Ciena

Atlanta, GA

- Diagnosed and resolved 30+ production defects across Waveserver OS CLI, telemetry, and performance-monitoring systems (C++/Python), including a regression undetected for over a year
- Resolved 25 Coverity-flagged null-pointer findings and led a test-coverage audit across 3 network interfaces, including YANG, OpenConfig, and SNMP, after root-causing CLI crash bugs

Software Engineering Intern

Aug 2025 – Sep 2025

4Earth

Kennesaw, GA

- Used Python to analyze 900,000+ soil chemistry and customer records, saving 40+ hours (25%) monthly
- Developed an online chemical calculator using HTML and an Android app to help users analyze soil inputs and improve decision-making, reducing human error by 20%

FPGA Engineering Intern

May 2024 – Aug 2024

Luon Energy

Johns Creek, GA

- Configured an FPGA-based subway train design in VHDL, enabling faster trains via optimized signal processing
- Built a breadboard-level railroad controller, increasing signal processing speed 15%

Teaching Assistant - Physics

Aug 2022 – Dec 2022

Kennesaw State University

Kennesaw, GA

- Conducted breakout sessions and guided students in grasping the fundamental principles of classical mechanics
- Directed test prep sessions for 20+ students, boosting class average by 10%

PROJECTS

Python Chess Engine

Jan 2026 – May 2026

- Built a Python chess engine with a PyGame GUI supporting all official chess rules
- Implemented minimax with alpha-beta pruning and heuristic board evaluation

Cybersecurity and Embedded Systems

Aug 2025 – Dec 2025

- Used ChipWhisperer to identify timing-based side-channel vulnerabilities via power analysis and fault injection
- Implemented RSA encryption/decryption in VHDL on a DE10 FPGA using modular exponentiation

Senior Capstone Design: Bandsaw Proximity Safety Sensor

Jan 2025 – May 2025

Georgia Institute of Technology

Atlanta, GA

- Designed and built a sensor to detect hand proximity to a bandsaw, improving engineer safety by 30%
- Implemented the system on a Raspberry Pi 5, providing real-time alerts when hands approach the saw

FPS Game Design

Jan 2024 – May 2024

- Assembled gaming hardware on a solderless breadboard, including a joystick, MCU (NXP LPC1768), speaker, and LCD screen from SparkFun
- Developed a multi-level LCD game in C++ featuring real-time jet control, dynamic environments, enemy behavior, projectile avoidance, and an integrated menu system

TECHNICAL SKILLS

Technical Languages: Java, SQL, RISC V Assembly, MIPS Assembly, C, C++, VHDL, Python, CUDA, Verilog, C#

Developer Tools: Quartus, Visual Studio, Mbed Studio, IntelliJ, Eclipse, Linux (Ubuntu)

Hardware Expertise: FPGAs, Oscilloscopes, Solderless breadboard, ARM Mbed microcontroller, ChipWhisperer

Spoken Languages: English (fluent), Mandarin (fluent), Hebrew (Beginner), Hindi (Beginner), Arabic (Beginner)

Extracurricular Activities: Chess Master, Violinist, Comparative Religion, 3D Printing, MMA, Speedcuber, Magician